

COVID_19 STATE-LEVEL DATA VISUALIZATIONS (REPORT)

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28th January 2026

MASTERS PROGRAM

GLOBAL PUBLIC HEALTH

SUBJECT: DIGITAL HEALTH

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LINKS:

GITHUB REPOSITORY

SHINY APPLICATION

Used Case

Public health problem

During COVID-19, large amounts of health, economic, demographic, and social data were collected. Stakeholders find it difficult to understand such raw data; that is why the need to convert that data into presentable forms arose to use it for public health decision makings. Interactive dashboards became popular which were produced after cleaning, analyzing, and exploring those data. Such visualizations provide clear, simple, and comparative information for decision makers and the public.

Main Objective

The primary objective was to translate COVID-19 data into an interactive dashboard. It should have easily understandable visualizations that could be easily interpreted, giving comparisons with other populations, and helping in decision making. These steps will help stakeholders in multiple ways. Raw datasets are not effective communication tools for non-technical audiences. Visualizations in the forms of graphs and charts help to highlight trends and patterns and support reliable public health communication.

This project is focusing on creating an interactive dashboard that will allow its users to explore COVID-19 data through visualizations rather than reading static reports. The project also aims to increase the understanding the equity in public health communication. The visualizations will support decision making, teaching, and public awareness about heath.

Stakeholders:

1. General public

General public users need a simple interface to understand how COVID-19 affected different regions, without advanced statistical knowledge. They always search for something easy to predict.

2. Educators

Teachers need something that could be used in classrooms to explain epidemiological concepts to students in an easy way. Visualizations on dashboards provide such ease.

3. Public health students and researchers

Higher studies in public health and research demand a learning tool to explore real COVID-19 data. Visualizations are needed to understand indicators such as infection and death rates, environmental factors affecting health, and public health trends.

4. Policy makers

They require summarized, comparable indicators (e.g., rates per population rather than raw counts) because they have to identify high-risk areas and allocate the resources more efficiently.

Data used in this project is taken from the CSV file called (COVID19_state.csv), which is also called raw data. It includes State name, Number of COVID-19 tests, infections, deaths, Population size, Income, ICU beds, Pollution levels, and other indicators.

SOLUTION

I have tried to create an interactive shiny application using COVID_19 data of United States of America available on the Kaggle website. This application is built considering epidemiological factors (infections and deaths), health system capacity (ICU beds), social factors (income, population), and environmental factors (pollution). The purpose was to explore the dependency of one factor on another. Such as: Relation of infection with number of deaths, ICU capacity in one state as compared to the other, pollutions levels in different states. These visualizations in the form of graphs and charts will provide these comparisons and impacts across the region of America.

Pathway to the Solution

Step 1:

R studio was used to develop shiny application because it is widely used in public health and it supports data cleaning and analysis, high-quality plots, and create Interactive dashboards.

Packages used in R studio for the project are:

- Pacman __ for package management
- shiny __ for building the interactive web application
- tidyverse __ for data manipulation and visualization
- readr __ for reading CSV files
- Dplyr __ for data transformation
- ggplot2 __ for creating plots
- janitor __ for cleaning variable names
- plotly __ for interactive visualizations

Step 2:

Instead of displaying raw case and death counts, the project focused on rates:

- Infection rate per 100,000 population
- Death rate per 100,000 population

This approach will give adequate comparisons between states with different population sizes.

Step 3:

Different charts and graphs were selected for different purposes:

- Scatter plot to identify infections and deaths rate

- Bar charts to compare state's ICU capacity
- Pie charts to show pollution contribution of states

Step 4: User friendly design

The dashboard was designed keeping non-technical users in mind:

- Simple labels
- Clear legends
- Minimal technical jargon
- Interactive filters (state selection, indicators)

Step 5: Improvements made

Initial plots were tested for clarity. Based on readability issues (e.g., overlapping labels), adjustments were made such as:

- Rotating axis labels
- Improving color contrast
- Adding tooltips instead of text overload

The final solution supports the rapid understanding of disease burden, teaching epidemiological concepts, and transparent communication of health data.

Data Cleaning and Preparation:

The raw was cleaned and analyzed before working on it. `Clean_names()` function was used to convert column names and remove spaces, to make data easier to work on R studio. Two indicators were derived using `mutate()` function:

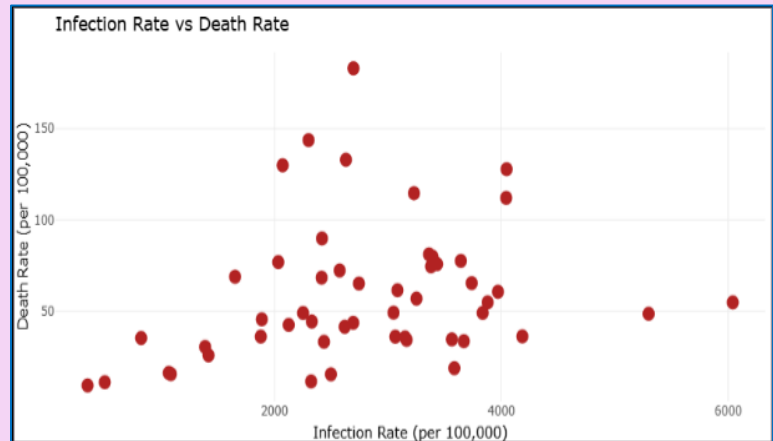
- **Infection rate:** number of infected cases per 100,000 population.
- **Death rate:** number of deaths per 100,000 population.

They made raw data meaningful and allowed easier comparison across the states. Limited number of variables were used to work on. `Drop_na()` function was used to remove missing values to make sure that visualizations are based on complete data. All these functions were used to make data set clear and well-organized, which was then called `covid_clean`.

Visualization Design:

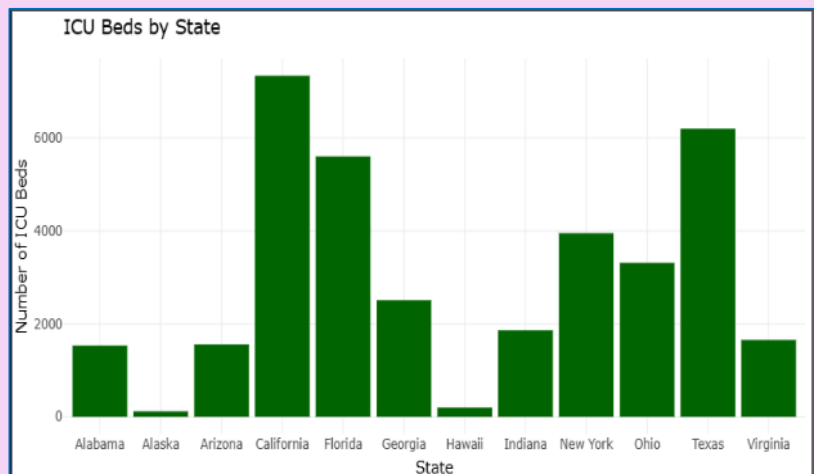
Tab 1: Infection vs Death (Scatter Plot)

This tab displays a scatter plot of infection vs death rate for the states. The **purpose** is to explore the widespreadness of COVID-19 infections and their mortality rates. States can be filtered based on **income range** and **population range** using slider inputs. Infection rate is on the x-axis and death rate is on the y-axis. We can hover over points and explore the data. This visualization helps to assess whether higher infection rates are associated with higher death rates and how this relationship changes across income and population levels.



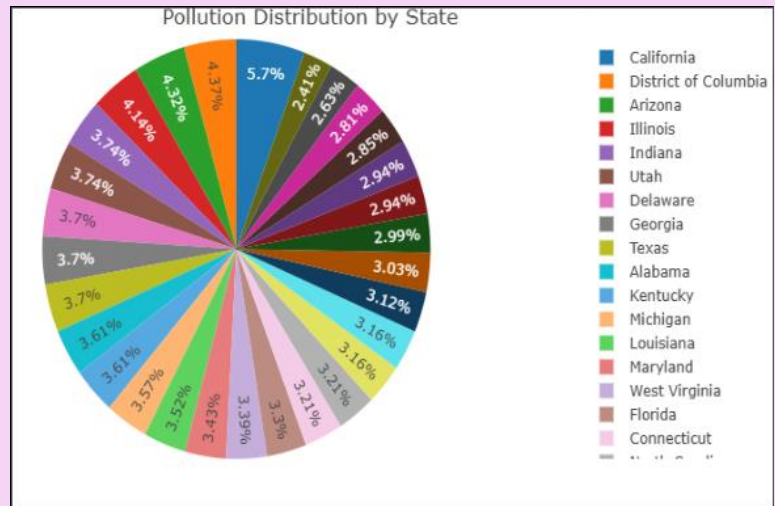
Tab 2: ICU Capacity (Bar Chart)

This tab shows a bar chart of ICU beds by state. The **purpose** is to compare ICU beds across states to help determine disparities in healthcare infrastructure. Specific states and their ICU bed capacity can be selected to determine the range. The resulting bar chart shows the number of ICU beds for the selected states. Hover over the bars for exact values. This shows differences in healthcare capacity in the states selected in filter area.



Tab 3: Pollution Pie Chart

This tab presents a pie chart showing the distribution of pollution levels in selected states. The **purpose** is to show that environmental factors such as pollution are potential contributors to respiratory issues, so, through this graph, pollution levels across regions can be visually compared. States and a pollution range can be filtered. Pollution values and percentage contribution of each state to total pollution among the selected group can be seen on the pie chart.



Implementation

Using R studio and working with R language as mentioned above, the projected was implemented as follows:

Shiny Dashboard Development

- **UI (User Interface):**
 - Dropdown menus for selecting state or indicator
 - Output panels for plots
- **Server logic:**
 - Reactive functions update plots when user changes selections
 - Ensures real-time interaction with the data

Publishing of shiny application

- The application was published online using **shinyapps.io** and can be run locally in RStudio)
- Users can access it through a web browser without installing R

The link is provided at the start of the document.

Supporting Files

- Dataset: COVID19_state.csv
- R script: app.R

Conclusion:

The project shows that an interactive Shiny dashboard can be used to explore COVID-19 data in multi-dimensional ways by combining health system capacity and environmental indicators instead of just focusing on case counts.
